

NET 621 - Week 6 BGP

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Week 6: BGP Fundamentals (Tasks 48-54)

Task 48-49: eBGP Configuration

Link: R3 G0/1 10.0.2.1/30 <-> R-ISP2 G0/0 10.0.2.2/30 Copper Cross Over. R-ISP2 Lo0 8.8.8.8/24.

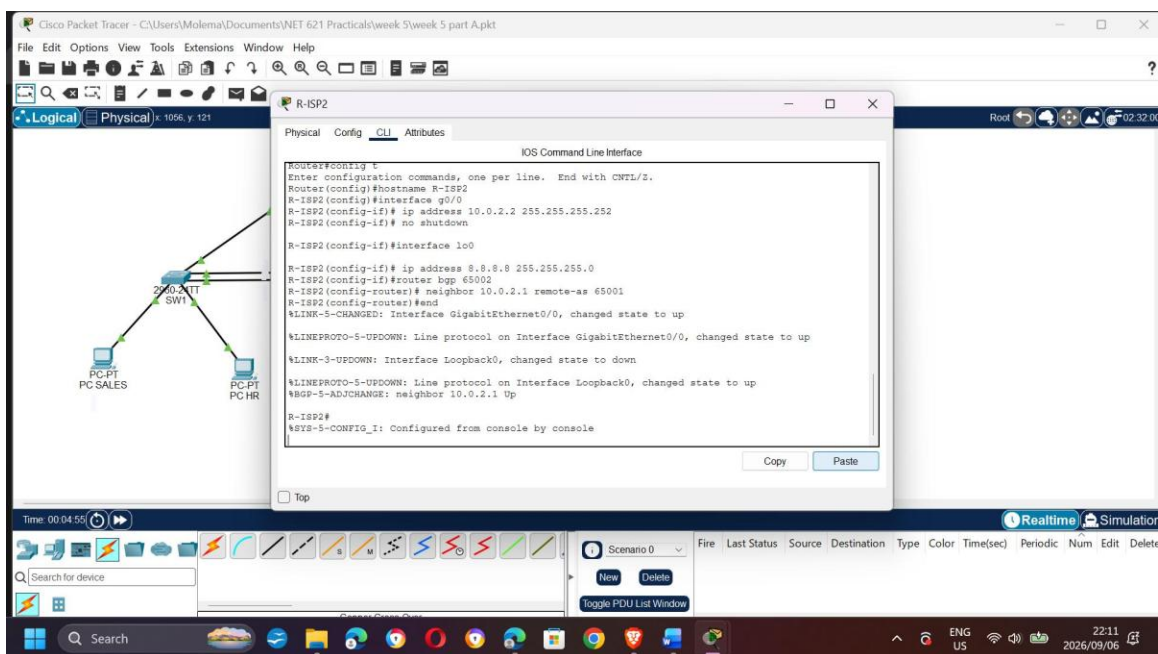


Figure 1: R-ISP2 eBGP config - hostname R-ISP2, interface g0/0 10.0.2.2 255.255.255.252 no shutdown, interface lo0 8.8.8.8 255.255.255.0, router bgp 65002 neighbor 10.0.2.1 remote-as 65001. Logs: LINK-5-CHANGED Gig0/0 up, LINEPROTO up, BGP-5-ADJCHANGE neighbor 10.0.2.1 Up - proof Task 49.

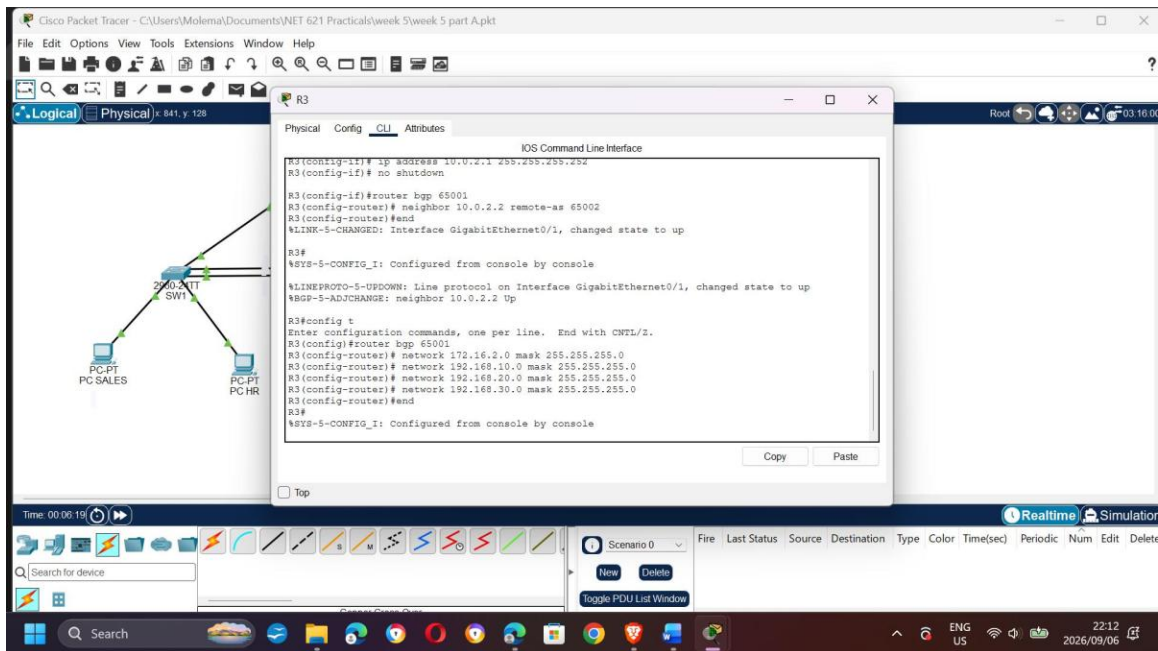


Figure 2: R3 eBGP config - interface g0/1 10.0.2.1 255.255.255.252, router bgp 65001 neighbor 10.0.2.2 remote-as 65002, LINK/ LINEPROTO up, BGP-5-ADJCHANGE neighbor 10.0.2.2 Up - proof Task 48. Then network statements: network 172.16.2.0 mask 255.255.255.0, network 192.168.10.0/20.0/30.0 mask - Task 50.

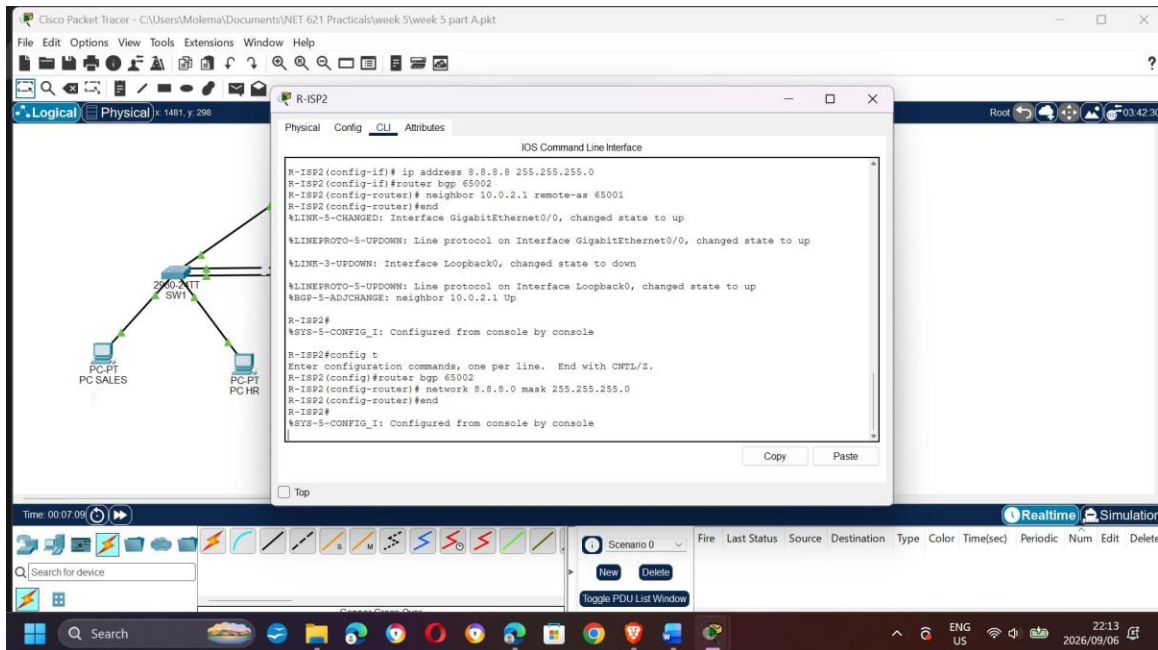


Figure 3: R-ISP2 advertising - router bgp 65002 network 8.8.8.0 mask 255.255.255.0 - Task 51.

Task 52: show ip bgp summary

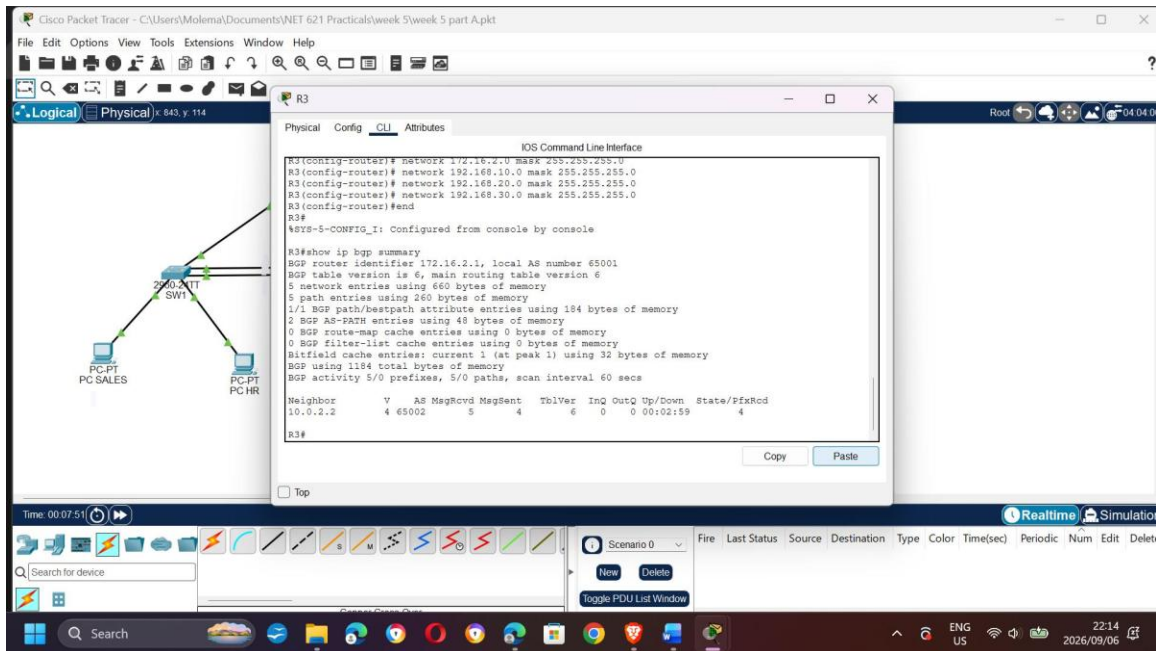


Figure 4: R3 show ip bgp summary - BGP router identifier 172.16.2.1 local AS 65001, 5 network entries, 5 path entries, Neighbor 10.0.2.2 V 4 AS 65002 MsgRcvd 5 MsgSent 4 TblVer 6 Up/Down 00:02:59 State/PfxRcd 4 - eBGP Up receiving 4 prefixes from ISP. Task 52 proof.

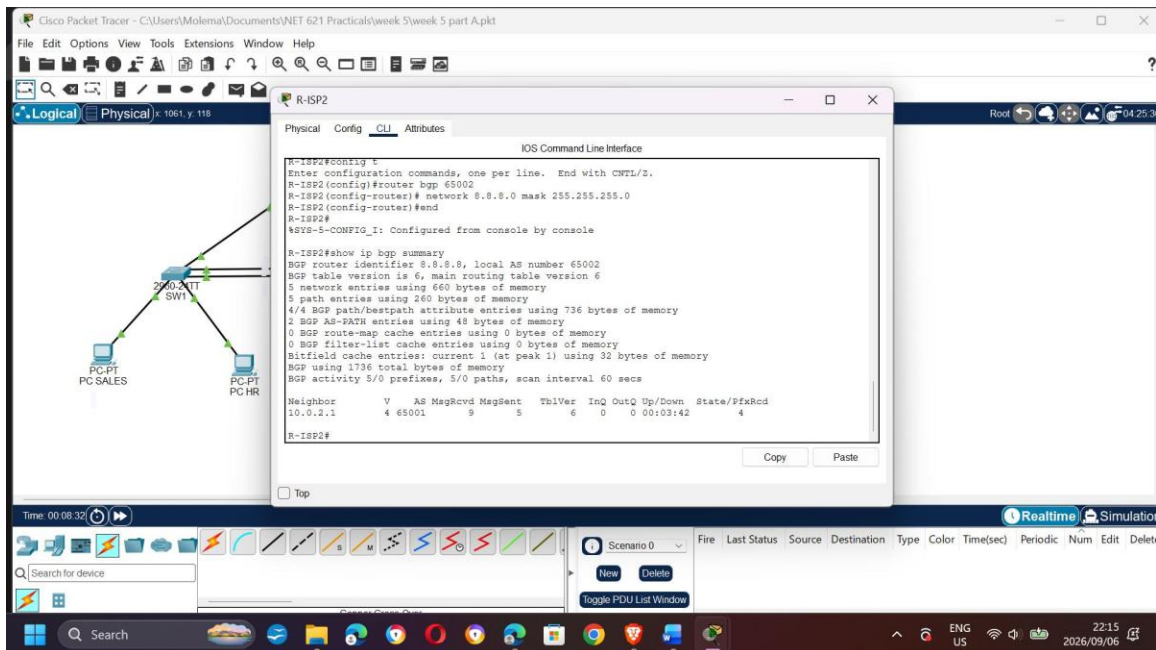


Figure 5: R-ISP2 show ip bgp summary - BGP router ID 8.8.8.8 local AS 65002, Neighbor 10.0.2.1 AS 65001 Up/Down 00:03:42 State/PfxRcd 4 - receives 4 enterprise prefixes. Task 52 both sides.

Task 53: show ip route bgp / show ip bgp

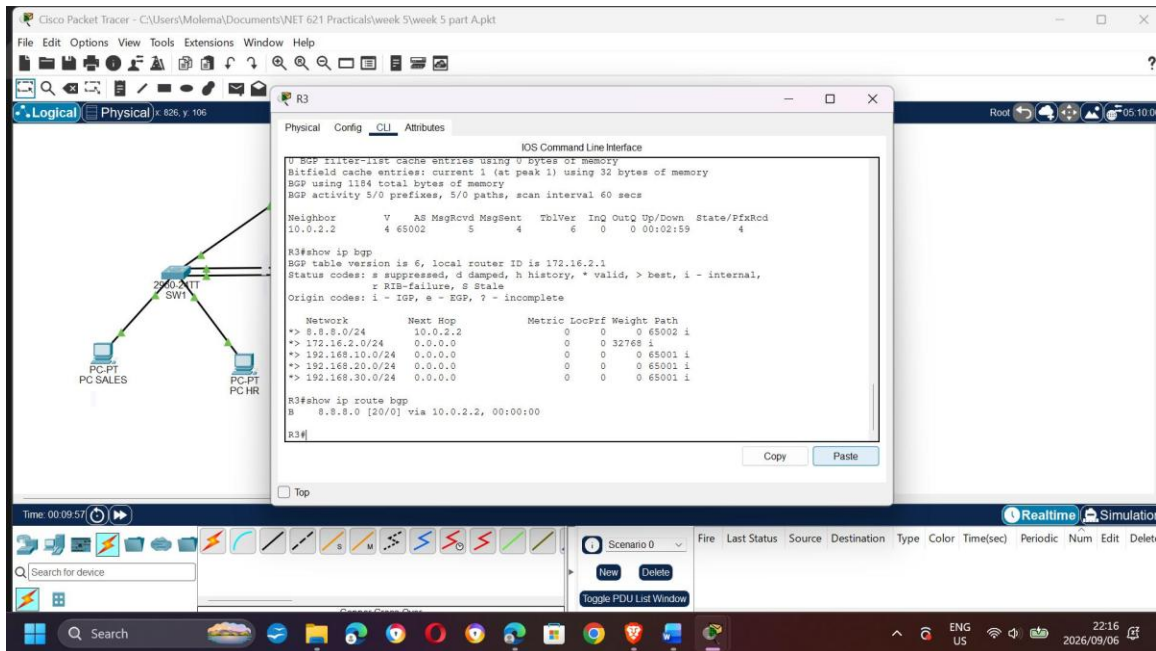


Figure 6: R3 show ip bgp - *> 8.8.8.0/24 Next Hop 10.0.2.2 Metric 0 Weight 0 Path 65002 i (learned via eBGP), *> 172.16.2.0/24 0.0.0.0, *> 192.168.10.0/24 etc. R3 show ip route bgp - B 8.8.8.0 [20/0] via 10.0.2.2, 00:00:00 - BGP route in RIB with AD 20. Task 53 proof. Note AD 20 better than OSPF 110.

Task 54: Loopback ping reachability

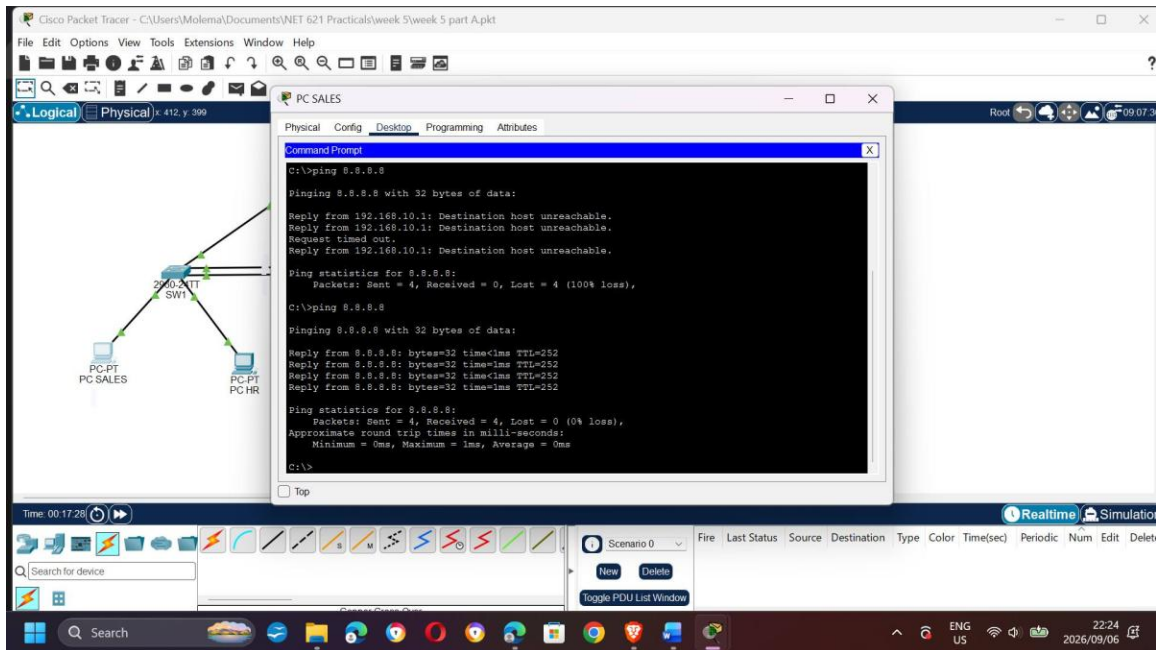


Figure 7: PC SALES (192.168.10.x) ping 8.8.8.8 - BEFORE redistribution: Reply from 192.168.10.1 Destination host unreachable 100% loss. AFTER redistribution (router ospf 1

redistribute bgp 65001 subnets + router bgp 65001 redistribute ospf 1): Reply from 8.8.8.8
bytes=32 time<1ms TTL=252 0% loss. TTL 252 = 255-3 hops R1->R-EDGE->R3->R-ISP2.
Proves end-to-end enterprise to ISP reachability via eBGP + redistribution. Task 54
complete.